GNS3 ENTERPRISE NETWORK INFRASTRUCTURE LAB

Portfolio Documentation

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Technologies Used:

GNS3 | VMware Workstation Pro | pfSense CE | Cisco IOS
Windows Server 2022 | Ubuntu Server 24.04 | Zabbix 7.0
Apache | Active Directory | OSPF | pfSense Firewall

1. Project Overview

This document details the design, implementation, and verification of a fully functional enterprise branch network lab built using GNS3 and VMware Workstation. The lab simulates a real-world enterprise network environment featuring perimeter security, dynamic routing, VLAN segmentation, centralized identity management, a DMZ web server, and comprehensive network monitoring.

The primary goal of this project is to demonstrate hands-on skills in network administration, system administration, and security — skills directly applicable to entry-level Network Administrator, System Administrator, and IT Support Engineer roles.

1.1 Objectives

- Design and implement an enterprise-grade network topology in GNS3
- Configure pfSense as a perimeter firewall with NAT and security rules
- Implement OSPF dynamic routing with MD5 authentication across three routers
- Configure VLAN segmentation with ACL-based inter-VLAN security policy
- Deploy Windows Server 2022 with Active Directory, DNS, and DHCP
- Set up an Ubuntu DMZ server running Apache web services
- Implement centralized monitoring using Zabbix 7.0 with email alerting
- Configure SSH access on all routers for secure remote management

1.2 Lab Environment

Component	Details
Host Machine	Physical PC, 16GB RAM
Hypervisor	VMware Workstation Pro 25
Network Simulator	GNS3 with GNS3 VM
Router Platform	Cisco 7200 Series (IOS)
Switch Platform	Cisco IOU L2
Firewall	pfSense CE 2.8.1
Server OS	Windows Server 2022
Linux OS	Ubuntu Server 24.04 LTS
Monitoring	Zabbix 7.0.25

2. Network Topology

The topology consists of a pfSense firewall at the edge, three Cisco 7200 routers forming the OSPF backbone, managed switches providing VLAN access, Windows Server 2022 as the enterprise domain controller, and Ubuntu Server in a dedicated DMZ.

2.1 Device Roles

Device	Type	Role
pfSenseFW	pfSense CE VM	Edge firewall, NAT, WAN gateway
HQ	Cisco 7200 Router	Head-office transit router, OSPF
BR2	Cisco 7200 Router	Branch core transit router, OSPF
BR1	Cisco 7200 Router	Branch gateway, inter-VLAN routing, DHCP relay
LANSW	Cisco IOU L2 Switch	LAN access switch, VLAN 10/20/99 trunk
DMZSW	Cisco IOU L2 Switch	DMZ access switch, VLAN 100
DC01	Windows Server 2022 VM	Domain Controller, DNS, DHCP
Ubuntu Server	Ubuntu 24.04 VM	DMZ web server, Apache, Zabbix

2.2 Physical Connections

From	Interface	To	Interface
pfSense Cloud	eth1 (VMnet2)	HQ	fa0/0
HQ	fa0/1	BR2	fa0/0
BR2	fa0/1	BR1	fa0/0
BR1	fa0/1	LANSW	e0/0
BR1	fa2/0	DMZSW	e0/0
LANSW	e0/1	PC1	e0
LANSW	e0/2	PC2	e0
LANSW	e1/1	PC3	e0
LANSW	e1/2	PC4	e0
LANSW	e2/0	DC01 Cloud	eth2 (VMnet3)
DMZSW	e0/1	Ubuntu Cloud	eth3 (VMnet4)

2.3 VMware Network Adapter Mapping

VMnet	Type	Purpose	Connected VM
VMnet0	Bridged	pfSense WAN — internet access	pfSense (Adapter 1)
VMnet2	Host-only	pfSense LAN to GNS3 HQ	pfSense (Adapter 2)
VMnet3	Host-only	DC01 to GNS3 LANSW	DC01 (Adapter 1)
VMnet4	Host-only	Ubuntu to GNS3 DMZSW	Ubuntu (Adapter 1)

3. IP Addressing Plan

3.1 Subnet Allocations

Network	Subnet	Purpose	Gateway
WAN (pfSense)	DHCP from ISP	pfSense WAN link	Physical network
HQ-pfSense	10.0.0.0/30	pfSense eth2 to HQ fa0/0	N/A
HQ-BR2	10.0.1.0/30	HQ fa0/1 to BR2 fa0/0	N/A
BR2-BR1	10.0.2.0/30	BR2 fa0/1 to BR1 fa0/0	N/A
VLAN 10 (HR)	192.168.10.0/24	PC1, PC2	192.168.10.1
VLAN 20 (IT)	192.168.20.0/24	PC3, PC4	192.168.20.1
VLAN 99 (MGMT)	192.168.99.0/24	DC01, Management	192.168.99.1
VLAN 100 (DMZ)	192.168.100.0/24	Ubuntu Server	192.168.100.1

3.2 Interface IP Assignments

Device	Interface	IP Address	Description
pfSense	em0 (WAN)	DHCP	Internet uplink
pfSense	em1 (LAN)	10.0.0.1/30	LAN to HQ
HQ	fa0/0	10.0.0.2/30	To pfSense
HQ	fa0/1	10.0.1.1/30	To BR2
BR2	fa0/0	10.0.1.2/30	To HQ
BR2	fa0/1	10.0.2.1/30	To BR1
BR1	fa0/0	10.0.2.2/30	To BR2
BR1	fa0/1.10	192.168.10.1/24	VLAN 10 gateway (HR)
BR1	fa0/1.20	192.168.20.1/24	VLAN 20 gateway (IT)
BR1	fa0/1.99	192.168.99.1/24	VLAN 99 gateway (MGMT)
BR1	fa2/0	192.168.100.1/24	VLAN 100 gateway (DMZ)
DC01	Ethernet	192.168.99.10/24	Static — Domain Controller
Ubuntu	ens32	192.168.100.10/24	Static — DMZ web server

3.3 DHCP Pools (served by DC01)

Scope	Range	Gateway	DNS Server
VLAN 10 — HR Users	192.168.10.50 —	192.168.10.1	192.168.99.10

Scope	Range	Gateway	DNS Server
	192.168.10.200		
VLAN 20 — IT Users	192.168.20.50 — 192.168.20.200	192.168.20.1	192.168.99.10

4. Router Configurations

4.1 HQ Router

Interface Configuration

```
interface FastEthernet0/0
 ip address 10.0.0.2 255.255.255.252
 ip ospf message-digest-key 1 md5 <password>
 ip ospf authentication message-digest
 no shutdown
!
interface FastEthernet0/1
 ip address 10.0.1.1 255.255.255.252
 ip ospf message-digest-key 1 md5 <password>
 ip ospf authentication message-digest
 no shutdown
```

OSPF Configuration

```
router ospf 1
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.1.0 0.0.0.3 area 0
 default-information originate always
```

Default Route

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

SSH Configuration

```
hostname HQ
ip domain-name lab.local
crypto key generate rsa modulus 1024
ip ssh version 2
username vasanth privilege 15 secret <password>
line vty 0 4
 transport input ssh
 login local
```

SNMP Configuration

```
snmp-server community VK_Lab@55 RO
snmp-server location GNS3-Lab
snmp-server contact vasanth@lab.local
```

4.2 BR2 Router

BR2 acts as the transit router between HQ and BR1. Configuration mirrors HQ with appropriate interface IPs.

Interface	IP Address	Connected To
fa0/0	10.0.1.2/30	HQ fa0/1
fa0/1	10.0.2.1/30	BR1 fa0/0

4.3 BR1 Router (Branch Gateway)

BR1 is the most complex router — it handles inter-VLAN routing via subinterfaces, DHCP relay to DC01, and DMZ connectivity.

Subinterface Configuration

```
interface FastEthernet0/1
 no ip address
 no shutdown
!
interface FastEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.99.10
!
interface FastEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.99.10
!
interface FastEthernet0/1.99
 encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
!
interface FastEthernet2/0
 ip address 192.168.100.1 255.255.255.0
 ip access-group DMZ_POLICY in
```

5. Switch Configurations

5.1 LANSW — LAN Access Switch

Interface	Mode	VLAN	Connected To
e0/0	Trunk	10, 20, 99	BR1 fa0/1
e0/1	Access	VLAN 10	PC1
e0/2	Access	VLAN 10	PC2
e1/1	Access	VLAN 20	PC3
e1/2	Access	VLAN 20	PC4
e2/0	Access	VLAN 99	DC01 (via cloud)

5.2 DMZSW — DMZ Access Switch

Interface	Mode	VLAN	Connected To
e0/0	Access	VLAN 100	BR1 fa2/0
e0/1	Access	VLAN 100	Ubuntu Server (via cloud)

6. pfSense Firewall Configuration

6.1 Interface Configuration

Interface	Physical NIC	IP	Role
WAN	em0	DHCP from physical network	Internet uplink
LAN	em1	10.0.0.1/30	Internal network gateway

6.2 Static Route

A static route is configured to direct all internal traffic back to HQ router:

Destination	Gateway	Interface	Purpose
192.168.0.0/16	10.0.0.2	LAN (em1)	Route to all internal VLANs via HQ

6.3 Outbound NAT Rules

Manual outbound NAT is configured to translate all internal traffic to the WAN IP:

Source	Destination	Translation	Description
192.168.0.0/16	Any	WAN Address	Internal VLANs to internet
10.0.0.0/8	Any	WAN Address	Transit links to internet

6.4 Firewall Rules

Rule	Action	Source	Destination	Purpose
1	Allow	Any	Any	Allow all internal traffic out (lab)
2	Allow	LAN net	DMZ 192.168.100.0/24	LAN to DMZ access
3	Block	DMZ net	LAN 192.168.0.0/16	DMZ cannot initiate to LAN

7. Security Policy (ACLs)

Access Control Lists are applied on BR1 to enforce inter-VLAN security. The policy follows a least-privilege model — each VLAN can only access what it needs.

7.1 VLAN Security Policy Summary

Source VLAN	Destination	Access	Reason
VLAN 10 — HR	VLAN 20 (IT)	BLOCKED	Department separation
VLAN 10 — HR	VLAN 100 (DMZ)	BLOCKED	HR has no need for DMZ
VLAN 10 — HR	VLAN 99 (MGMT)	ALLOWED	Needs AD/DNS/DHCP
VLAN 10 — HR	Internet	ALLOWED	Normal browsing
VLAN 20 — IT	VLAN 10 (HR)	BLOCKED	IT cannot access HR data
VLAN 20 — IT	VLAN 100 (DMZ)	ALLOWED	IT manages DMZ servers
VLAN 20 — IT	Internet	ALLOWED	Normal browsing
VLAN 99 — MGMT	Everywhere	ALLOWED	Admin VLAN — full access
VLAN 100 — DMZ	VLAN 20 (IT)	ALLOWED	Respond to IT requests
VLAN 100 — DMZ	VLAN 99 (MGMT)	ALLOWED	Respond to admin requests
VLAN 100 — DMZ	VLAN 10 (HR)	BLOCKED	DMZ cannot reach HR
VLAN 100 — DMZ	Internet	ALLOWED	Updates and outbound

7.2 ACL Configurations on BR1

VLAN10_POLICY (applied inbound on fa0/1.10)

```
ip access-list extended VLAN10_POLICY
 deny   ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny   ip 192.168.10.0 0.0.0.255 192.168.100.0 0.0.0.255
 permit ip any any
```

VLAN20_POLICY (applied inbound on fa0/1.20)

```
ip access-list extended VLAN20_POLICY
 deny   ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 permit ip any any
```

DMZ_POLICY (applied inbound on fa2/0)

```
ip access-list extended DMZ_POLICY
 permit ip 192.168.100.0 0.0.0.255 192.168.20.0 0.0.0.255
 permit ip 192.168.100.0 0.0.0.255 192.168.99.0 0.0.0.255
```

```
deny   ip 192.168.100.0 0.0.0.255 192.168.10.0 0.0.0.255
deny   ip 192.168.100.0 0.0.0.255 192.168.0.0 0.16.0.0
permit ip any any
```

8. Windows Server 2022 (DC01)

8.1 Server Configuration

Setting	Value
Hostname	DC01
IP Address	192.168.99.10/24
Default Gateway	192.168.99.1 (BR1)
DNS	127.0.0.1 (self)
Domain	lab.local
VLAN	99 (Management)

8.2 Active Directory

DC01 is configured as the primary Domain Controller for the lab.local domain. The following organizational structure has been implemented:

- Domain: lab.local
- Organizational Unit: IT DEPARTMENT
- Security Group: IT_Staff
- Domain Users: vasanth, Ben Stokes, Dhanush, Gokul, Madesh, Sandeep, Vignesh
- Domain Computer: CLIENT1

8.3 DHCP Scopes

Scope Name	Network	Range	Gateway	DNS
VLAN10-Users	192.168.10.0/24	10.50 — 10.200	192.168.10.1	192.168.99.10
VLAN20-Users	192.168.20.0/24	20.50 — 20.200	192.168.20.1	192.168.99.10

8.4 PowerShell Automation Scripts

The following PowerShell scripts are stored in C:\Scripts\ for AD administration:

Script	Purpose
BulkCreateUsers.ps1	Bulk create AD users from CSV file
ADUserReport.ps1	Generate report of all AD users
AddUsersToGroup.ps1	Add multiple users to a security group
InactiveUsers.ps1	Find and disable inactive user accounts
UnlockAccounts.ps1	Unlock locked-out user accounts

Script	Purpose
ServiceCheck.ps1	Check status of critical services
UpdateDepartment.ps1	Bulk update user department attributes

9. Ubuntu Server 24.04 (DMZ)

9.1 Server Configuration

Setting	Value
Hostname	ubuntu-server
IP Address	192.168.100.10/24
Default Gateway	192.168.100.1 (BR1 fa2/0)
DNS Servers	192.168.99.10, 8.8.8.8
VLAN	100 (DMZ)
Interface	ens32

9.2 Network Configuration (netplan)

```
network:
  version: 2
  ethernets:
    ens32:
      dhcp4: no
      addresses:
        - 192.168.100.10/24
      routes:
        - to: default
          via: 192.168.100.1
      nameservers:
        addresses:
          - 192.168.99.10
          - 8.8.8.8
```

9.3 Apache Web Server

Apache2 is installed and serves a custom corporate DMZ web page at http://192.168.100.10. The page displays server hostname, IP address, VLAN assignment, and admin details.

- Service: apache2
- Port: 80 (HTTP)
- Document root: /var/www/html/
- Status: enabled and running on boot

9.4 Zabbix Monitoring Stack

Ubuntu hosts the complete Zabbix 7.0 monitoring stack including server, frontend, and agent:

Component	Version	Port	Purpose
Zabbix Server	7.0.25	10051	Monitoring engine
Zabbix Frontend	7.0.25	80/443	Web dashboard (Apache)
Zabbix Agent	7.0.25	10050	Local host monitoring
MySQL	8.0	3306	Zabbix database backend

10. Zabbix Monitoring Configuration

10.1 Monitored Hosts

Host	Template	Interface	IP:Port
Zabbix server	Linux by Zabbix agent, Zabbix server health	Agent	127.0.0.1:10050
Ubuntu-Server	Linux by Zabbix agent, ICMP Ping	Agent	192.168.100.10:10050
DC01	Windows by Zabbix agent, ICMP Ping	Agent	192.168.99.10:10050
HQ-Router	Cisco IOS by SNMP	SNMP	10.0.0.2:161
BR2-Router	Cisco IOS by SNMP	SNMP	10.0.1.2:161
BR1-Router	Cisco IOS by SNMP	SNMP	10.0.2.2:161

10.2 SNMP Configuration

All routers are configured with a custom SNMP community string for security:

```
snmp-server community VK_Lab@55 RO
snmp-server location GNS3-Lab
snmp-server contact vasanth@lab.local
```

10.3 Email Alerting

Setting	Value
SMTP Server	smtp.gmail.com
SMTP Port	587
Security	STARTTLS
Alert Email	vasanthkumarvk2855@gmail.com
Minimum Severity	Warning
Recovery Alerts	Enabled

10.4 Alert Types Configured

- Host down / unreachable
- Router restart detected (uptime < 10 minutes)
- High memory utilization (> 90% for 5 minutes)
- Apache web service down
- Zabbix agent unavailable

- Interface high error rate
- Recovery notifications when issues resolve

11. Verification and Testing

11.1 OSPF Verification

OSPF neighborship was verified on all routers. All neighbors achieved FULL state with MD5 authentication:

Router	Neighbor	State	Interface
HQ (1.1.1.1)	BR2 (2.2.2.2)	FULL/BDR	fa0/1
BR2 (2.2.2.2)	HQ (1.1.1.1)	FULL/BDR	fa0/0
BR2 (2.2.2.2)	BR1 (3.3.3.3)	FULL/DR	fa0/1
BR1 (3.3.3.3)	BR2 (2.2.2.2)	FULL/DR	fa0/0

11.2 End-to-End Connectivity Matrix

Source	Destination	Expected	Result
PC1 (VLAN10)	192.168.10.1 (BR1 GW)	PASS	PASS
PC1 (VLAN10)	192.168.99.10 (DC01)	PASS	PASS
PC1 (VLAN10)	192.168.20.50 (PC3)	BLOCK	BLOCK
PC1 (VLAN10)	192.168.100.10 (Ubuntu)	BLOCK	BLOCK
PC1 (VLAN10)	8.8.8.8 (Internet)	PASS	PASS
PC3 (VLAN20)	192.168.100.10 (Ubuntu)	PASS	PASS
PC3 (VLAN20)	192.168.10.51 (PC1)	BLOCK	BLOCK
Ubuntu (DMZ)	192.168.99.10 (DC01)	PASS	PASS
Ubuntu (DMZ)	192.168.10.51 (PC1)	BLOCK	BLOCK
Ubuntu (DMZ)	8.8.8.8 (Internet)	PASS	PASS
DC01 (MGMT)	192.168.100.10 (Ubuntu)	PASS	PASS
DC01 (MGMT)	8.8.8.8 (Internet)	PASS	PASS

11.3 DHCP Verification

All PCs successfully received IP addresses from DC01 via DHCP relay on BR1:

PC	IP Received	VLAN	DHCP Server
PC1	192.168.10.51/24	VLAN 10	192.168.99.10 (DC01)
PC2	192.168.10.52/24	VLAN 10	192.168.99.10 (DC01)
PC3	192.168.20.50/24	VLAN 20	192.168.99.10 (DC01)

PC	IP Received	VLAN	DHCP Server
PC4	192.168.20.51/24	VLAN 20	192.168.99.10 (DC01)

11.4 Service Verification

Service	Host	Test Method	Status
Active Directory	DC01	Domain join, user auth	Operational
DNS	DC01	nslookup lab.local	Operational
DHCP	DC01	ip dhcp on VPCS	Operational
Apache Web Server	Ubuntu	http://192.168.100.10	Operational
SSH (HQ)	10.0.0.2	PuTTY from DC01	Operational
SSH (BR2)	10.0.1.2	PuTTY from DC01	Operational
SSH (BR1)	10.0.2.2	PuTTY from DC01	Operational
Zabbix Monitoring	Ubuntu	http://192.168.100.10/ zabbix	Operational
NAT/Internet	pfSense	ping 8.8.8.8 from all hosts	Operational

12. Known Issues and Limitations

Issue	Impact	Workaround
GNS3 cable instability on restart	Some router links drop after GNS3 VM restart	Redraw affected cables after each restart
Zabbix and Apache on same host	Zabbix WebUI unavailable when Apache is stopped	Architectural limitation in single-VM setup
pfSense on bridged network	WAN IP changes with physical network	Lab environment limitation; static WAN not needed

13. Skills Demonstrated

Networking

- Enterprise network design and topology planning
- OSPF dynamic routing with MD5 authentication
- VLAN configuration and 802.1Q trunking
- Inter-VLAN routing using router-on-a-stick
- ACL design and implementation
- NAT/PAT configuration on pfSense
- DHCP relay using ip helper-address
- SSH hardening on Cisco routers
- SNMP monitoring configuration

System Administration

- Active Directory Domain Services deployment
- DNS and DHCP server configuration on Windows Server 2022
- PowerShell scripting for AD automation
- Ubuntu Server administration and service management
- Apache web server deployment and configuration
- Netplan network configuration on Ubuntu

Security

- pfSense perimeter firewall configuration
- DMZ design and implementation
- VLAN-based network segmentation
- ACL-based access control policy
- OSPF MD5 authentication
- SSH key-based authentication on routers

Monitoring & Operations

- Zabbix 7.0 deployment with MySQL backend
- Multi-platform monitoring (Linux, Windows, Cisco SNMP)
- Email alert configuration with Gmail SMTP
- Custom trigger creation for service monitoring
- Dashboard creation for infrastructure overview

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